

Elie Rouphael

Postdoctoral Researcher in AI, System Identification & Control

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Research Interests

Stochastic realization theory, hybrid system identification, probabilistic modeling, digital twins, vehicle dynamics modeling, time series learning, probabilistic modeling, Bayesian inference, deep state-space models, AI.

Education

PhD in Automatic Control and Computer Science

2021–2025

University of Lille – CRIStAL Laboratory, France

Thesis: Towards Stochastic Realization Theory for Linear Switched Models

Focus: Stochastic hybrid system identification, minimal realizations, learning with uncertainty.

Master's in Automatic Control and Electrical Energy

2020–2021

University of Poitiers – LIAS Laboratory, France

Graduated top of class. Research internship on non-stationary sinusoidal disturbance estimation.

Engineering Degree in Electrical and Industrial Control Systems

2016–2021

Lebanese University – Faculty of Engineering, Lebanon

Professional Experience

Postdoctoral Researcher – LabCom I-TireLab

Nov 2025 – Present

Working, with MICHELIN, on real-time prediction of slippery scenarios using hybrid modeling and machine learning, with the goal of improving vehicle safety. Developing digital-twin-based methods for tire-road interaction modeling and dynamics prediction.

PhD Researcher

Sep 2021 – Oct 2025

University of Lille – CRIStAL Laboratory

- Developed a theoretical framework for stochastic realization of linear switched systems.
- Designed system identification algorithms accounting for dynamical switching.
- Worked with SZTAKI (Budapest) on deep time series modeling with generalization guarantees.

ATER – Teaching and Research Associate – University of Lille

Sep 2024 – 2025

- 195+ hours of teaching: Python, C, Control Theory, Microcontrollers, GRAFCET.
- Supervising 3 student internships and 1 bachelor's project in machine learning and simulation.

Teaching Assistant (DCACE) – University of Lille

Oct 2022 – Sep 2024

128+ hours of tutorials and labs in automation, signal processing, and feedback systems.

Research Intern – LIAS Laboratory, University of Poitiers

Mar – Jul 2021

Topic: Detection of non-stationary harmonic disturbances in electrical systems.

Engineering Intern – Cimenterie Nationale, Lebanon

Jul – Aug 2019

Installation and commissioning of electrical control panels in industrial systems.

Certifications & Courses

IBM Certificates: - Introduction to DevOps - Deep Learning with Keras - Data Engineering - Machine Learning with Python - Python for AI & Development - Python Project for Data Engineering — **UCSC: Bayesian Statistics** — **DeepLearning.ai: Neural Networks and Deep Learning** — **EECI Course: Sparsity & Big Data in Control/ML** — **Spring School: Closed-loop & nonlinear system ID and Deep Learning** — **ACES Summer School: 60h in modern control theory**

Selected Projects (Supervised)

- **PAC Bounds for Stable SSMs:** This project studies Probably Approximately Correct (PAC) generalization bounds for deep advanced structures of State Space Models (SSMs). Collaborating with the SZTAKI research institute in Budapest, the project aims to theoretically justify the generalization ability of these models when they are stable.
- **Comparison of Classical and Deep Learning vs SSMs Time Series Models:** This project benchmarks traditional models (e.g., ARIMA, ARMAX, RNNs, CNNs, and Transformers) against SSMs (e.g., MAMBA) for time series forecasting.
- **Transport Mode Prediction with Hauts-de-France Data:** This project predicts individual transport mode choices (e.g., car, train, bike) to optimize mobility planning. It improves economic efficiency and ecological impact, by helping policymakers promote sustainable and cost-effective transportation strategies.
- **System Identification toolbox:** An ongoing initiative to develop and contribute to an open-source toolbox for system identification, focusing on LPV, hybrid, and switched systems.

Publications

- **Rouphael, E.**, et al. *Minimal covariance realization and system identification algorithm for a class of stochastic linear switched systems with i.i.d. switching*. IEEE CDC 2024.
- **Rouphael, E.**, et al. *Toward Stochastic Realization Theory for Generalized Linear Switched Systems With Inputs: Decomposition Into Stochastic and Deterministic Components and Existence and Uniqueness of Innovation Form*. IEEE L-CSS, 2024.
- **Rouphael, E.**, et al. *Variable frequency monitoring of power grid: a modified observer approach*. IFAC-PapersOnLine, 2023.
- **Rouphael, E.**, et al. *On minimal LPV state-space representations in innovation form: an algebraic characterization*. IEEE CDC 2022.

Technical Skills

Programming: Python, MATLAB, C/C++, SQL, R

Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy

Teaching (labs, projects): Python, C, Control Theory, Systems & Signals, Machine Learning

Statistical Tools: Time series statistics, probabilistic modeling, Bayesian inference

Expertise: System Identification, Hybrid Systems, Time Series Forecasting, Uncertainty Modeling, Deep Learning

Languages

Arabic (Native), French (DELFI – B2), English (TOEIC 865 / 990)

References

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University of Lille

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Dr. Mihály Petreczky

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